

Module 4 — Core Skills Drill: Descriptive Analytics

Write three functions that compute summary statistics, create distribution plots, and build a correlation heatmap. This drill targets retrieval practice — write the code from memory, not by copying from the reading.

Due: Tonight (Day 1, after the concept session). Resubmissions are accepted through Saturday of the assignment week.

Setup

1. **Accept the assignment:** [Accept the M4 Drill assignment](#)
2. Clone your repo and open it in VS Code.
3. The repo contains `data/sample_sales.csv` — a small dataset (~200 rows) with columns: `date`, `product`, `category`, `quantity`, `unit_price`, `region`, `revenue`, and `discount`. The four numeric columns are `quantity`, `unit_price`, `revenue`, and `discount`.
4. Install dependencies: `pip install -r requirements.txt` This installs `seaborn` and `scipy` (new packages for this module) along with `pandas`, `numpy`, `matplotlib`, and `pytest`.

Tasks

Complete the starter file `drill_eda.py`. The script should load the CSV, call all three functions, and save outputs to the `output/` directory. Include an `if __name__ == "__main__":` guard.

Task 1 — Summary Statistics: `compute_summary`

Write a function `compute_summary(df)` that:

- Takes a `DataFrame` as input
- Returns a summary `DataFrame` containing **count**, **mean**, **median**, **std**, **min**, **max** for all numeric columns
- Saves the summary to `output/summary.csv`

What the autograder checks: `output/summary.csv` exists, contains rows for `mean`, `std`, `min`, and `max`, and covers all four numeric columns.

Task 2 — Distribution Plots: `plot_distributions`

Write a function `plot_distributions(df, columns, output_path)` that:

- Takes a `DataFrame`, a list of column names, and an output file path
- Creates a **2x2 subplot figure** with histograms (using `sns.histplot` with KDE overlay) for the specified columns
- Adds a title to each subplot
- Saves the figure to the specified path (e.g., `output/distributions.png`)

Call this function with the four numeric columns: `quantity`, `unit_price`, `revenue`, and `discount`.

What the autograder checks: `output/distributions.png` exists and is a real plot (not an empty file).

Task 3 — Correlation Heatmap: `plot_correlation`

Write a function `plot_correlation(df, output_path)` that:

- Takes a `DataFrame` and an output file path
- Computes the **Pearson correlation matrix** for all numeric columns
- Visualizes it as an **annotated Seaborn heatmap** with a `coolwarm` color map
- Saves the figure to the specified path (e.g., `output/correlation.png`)

What the autograder checks: `output/correlation.png` exists and is a real plot (not an empty file).

Submit

1. Create branch `drill-4-descriptive-analytics`
2. Add `drill_eda.py` and the `output/` directory with all generated files
3. Commit, push, and open a PR to `main`
4. The autograder will run and report pass/fail
5. Paste your PR URL into TalentLMS → Module 4 → Core Skills Drill